

Real Analysis HW 8

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23rd August 2026

BS - Page 144 Exercise 8

Prove that if f and g are each uniformly continuous on $\mathbb{R}$, then the composite function $f \circ g$ is uniformly continuous on $\mathbb{R}$

Since f, g are two uniformly continuous functions $\forall \epsilon > 0 \exists \delta_0, \delta_1$ s.t. (with each δ independent of c)

$$|x - c| < \delta_0 \Rightarrow |g(x) - g(c)| < \epsilon$$

and

$$|x - c| < \delta_1 \Rightarrow |f(x) - f(c)| < \epsilon$$

For g , picking its $\epsilon = \delta_1$, we get the chain

$$|x - c| < \delta_0 \Rightarrow |g(x) - g(c)| < \delta_1 \Rightarrow |f(g(x)) - f(g(c))| < \epsilon$$

Thus $f \circ g$ is uniformly continuous, since the choice of both deltas were independent of c

BS - Page 167 Exercise 7

Suppose that $f : \mathbb{R} \rightarrow \mathbb{R}$ is differentiable at c and that $f(c) = 0$. Show that $g(x) := |f(x)|$ is differentiable at c if and only if $f'(c) = 0$

($\Rightarrow$)

Suppose g differentiable at c ,

Suppose FTSOC $f'(c) \neq 0$,

$\Rightarrow f(c)$ is not a relative minimum or maximum

$$\Rightarrow \forall \delta > 0 \ |x - c| < \delta \Rightarrow f(c) > f(x) \text{ and } f(c) < f(x)$$

Thus in some delta interval around c , f attains both positive and negative values, since $f(c) = 0$

Also we have

$$g'(c) = \lim_{x \rightarrow c} \frac{g(x) - g(c)}{x - c} = \lim_{x \rightarrow c} \frac{|f(x)|}{x - c}$$

Then choosing sequences which converge to c from the left and the right, say $x_n = c - \frac{1}{2n}$ and $y_n = c + \frac{1}{2n}$, we have that $x_n - c < \delta \Rightarrow f(x_n) < 0$ or $y_n - c < \delta \Rightarrow f(y_n) < 0$, WLOG assume the first case

$$x_n - c < \delta \Rightarrow f(x_n) < 0 \Rightarrow |f(x_n)| = -f(x_n)$$

$$\Rightarrow g'(c) = \lim_{n \rightarrow \infty} \frac{-f(x_n)}{x_n - c} = -f'(c)$$

Then

$$y_n - c < \delta \Rightarrow f(y_n) > 0 \Rightarrow |f(y_n)| = f(y_n)$$

$$\Rightarrow g'(c) = \lim_{n \rightarrow \infty} \frac{f(y_n)}{y_n - c} = f'(c)$$

Hence $f'(c) = -f'(c)$, which is a contradiction since we assumed $f'(c) \neq 0$

($\Leftarrow$)

Suppose $f'(c) = 0$

Then

$$f'(c) = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} = \lim_{x \rightarrow c} \frac{f(x)}{x - c} = 0$$

Also

$$g'(c) = \lim_{x \rightarrow c} \frac{|f(x)|}{x - c}$$

Then

$$|g'(c)| = \lim_{x \rightarrow c} \left| \frac{|f(x)|}{x - c} \right| = \lim_{x \rightarrow c} \left| \frac{f(x)}{x - c} \right| = |f'(c)| = 0$$

Hence $g'(c) = 0$

BS - Page 167 Exercise 9

Prove that if $f : \mathbb{R} \rightarrow \mathbb{R}$ is an even function, and has a derivative at every point, then the derivative f' is an odd function. Also prove that if $g : \mathbb{R} \rightarrow \mathbb{R}$ is an odd differentiable function then g' is an even function

(a)

We have that $f(-x) = f(x)$,

$$(f(x))' = (f(-x))'$$

Then by the Chain Rule

$$f'(x) = f'(-x)(-1)$$

$$\Rightarrow f'(-x) = -f'(x)$$

Thus $f'(x)$ is an odd function

(b)

We have that $g(-x) = -g(x)$,

$$(g(-x))' = (-g(x))'$$

Then by the Chain Rule

$$g'(-x)(-1) = -g'(x)$$

$$\Rightarrow g'(-x) = g(x)$$

Thus $g'(x)$ is an even function

BS - Page 175 Exercise 11

Give an example of a uniformly continuous function on $[0, 1]$ that is differentiable on $(0, 1)$, but whose derivative is not bounded on $(0, 1)$

Consider $f(x) = \sqrt{x}$, which I will prove is continuous on $[0, 1]$

Let $\epsilon > 0$

Choose $\delta = \epsilon^2$

Then for $|x - c| < \delta$

$$\begin{aligned} |f(x) - f(c)|^2 &= |\sqrt{x} - \sqrt{c}|^2 \\ &\leq |\sqrt{x} - \sqrt{c}| |\sqrt{x} + \sqrt{c}| \\ &< |x - c| \\ &< \delta \\ &< \epsilon^2 \end{aligned}$$

Thus

$$|f(x) - f(c)| < \epsilon$$

Since our choice of δ was independent of c , $f(x)$ is uniformly continuous

Now we look at $f'(x) = \frac{1}{2\sqrt{x}}$

Let $B > 0$

Assume $f'(x)$ is bounded

Then for all $x \in (0, 1)$, we have

$$\begin{aligned}\frac{1}{2\sqrt{x}} &\leq B \\ \Rightarrow \sqrt{x} &\geq \frac{1}{2B}\end{aligned}$$

But since $\lim_{x \rightarrow 0} \sqrt{x} = 0$, we get that it can be arbitrarily small, thus for any given $B > 0$, $\exists x$ s.t. the following inequality doesn't hold

$$\sqrt{x} \geq \frac{1}{2B} > 0$$

A contradiction, thus $f'(x)$ is unbounded